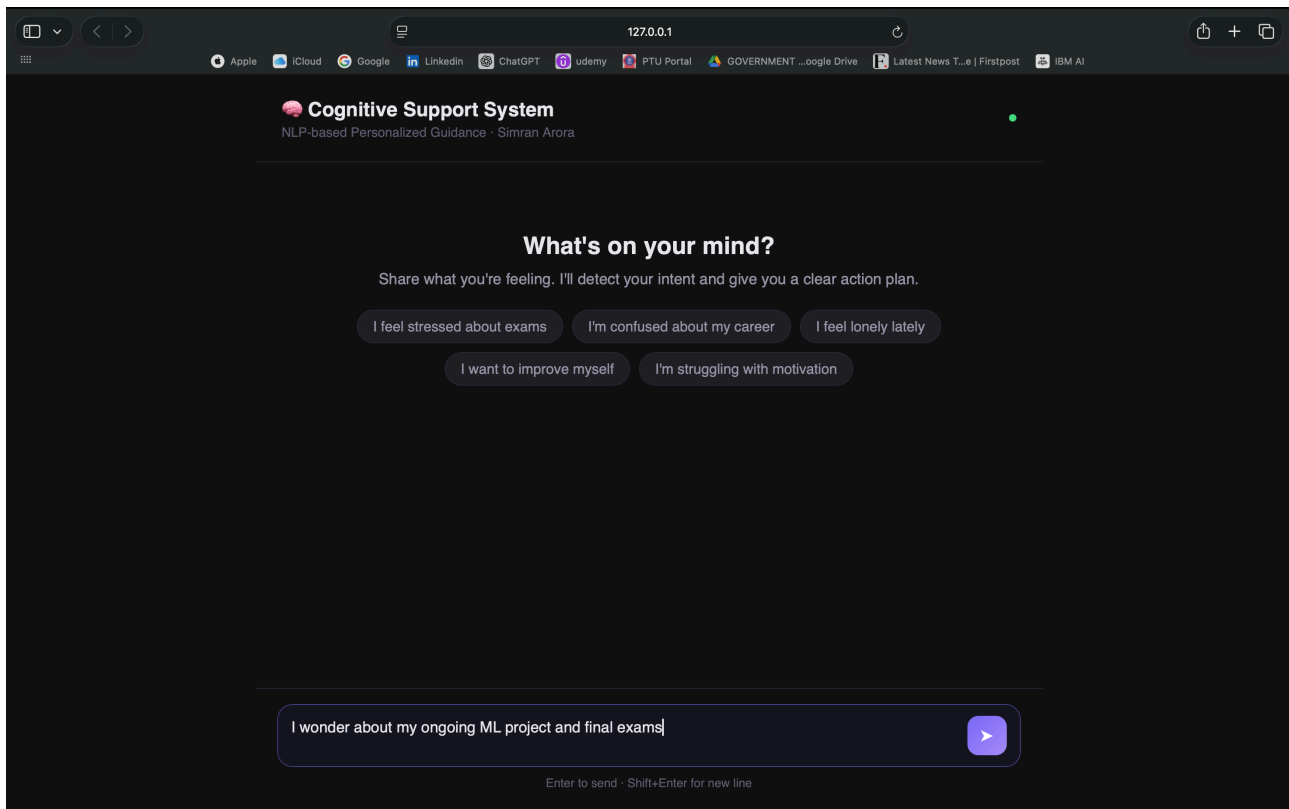
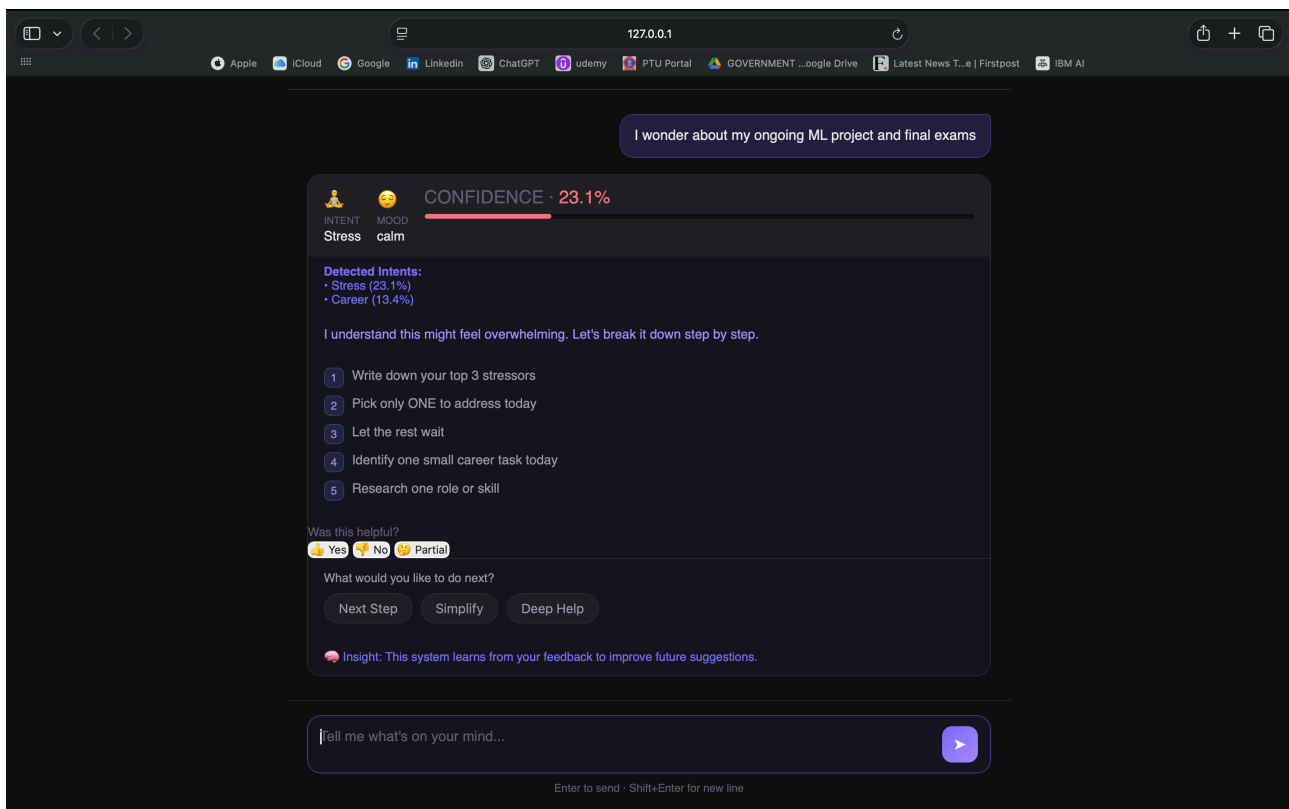


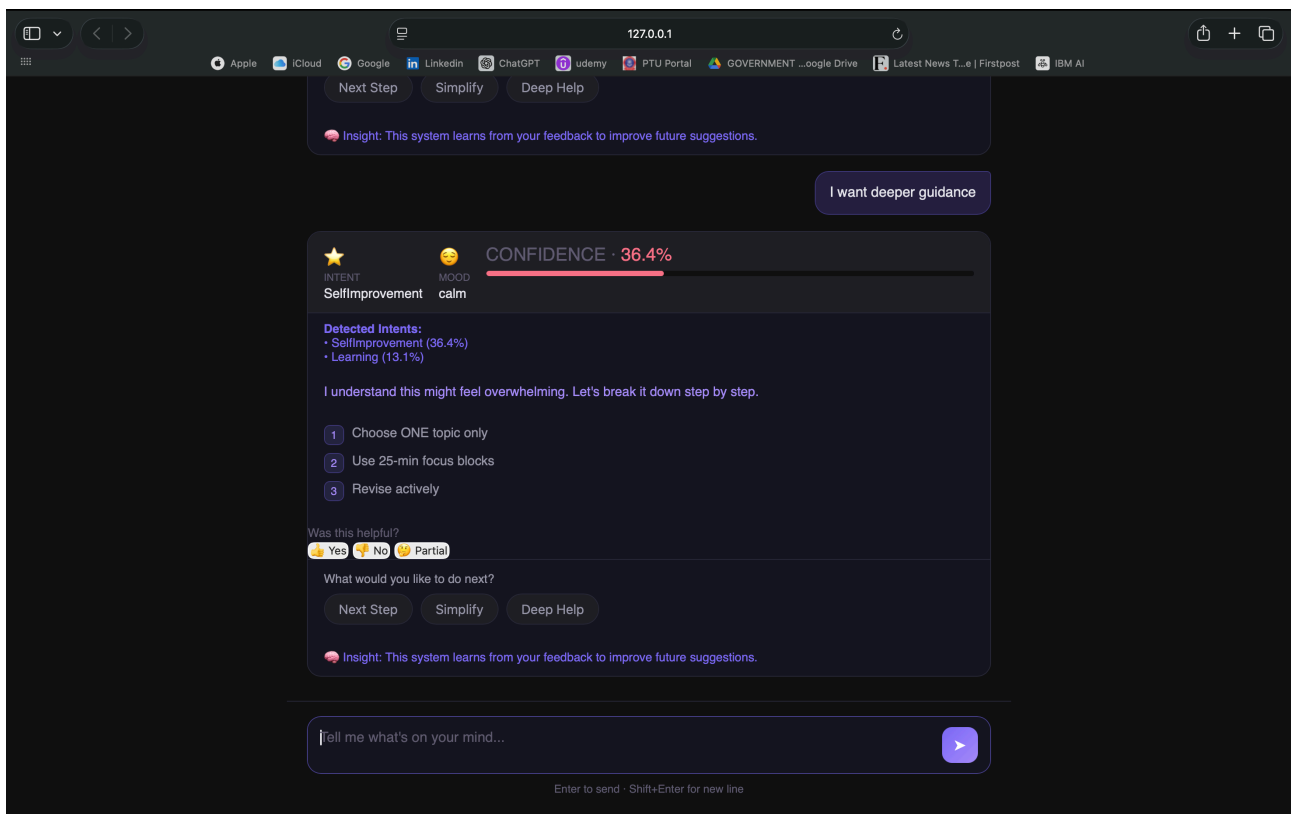
Implementation/Working:



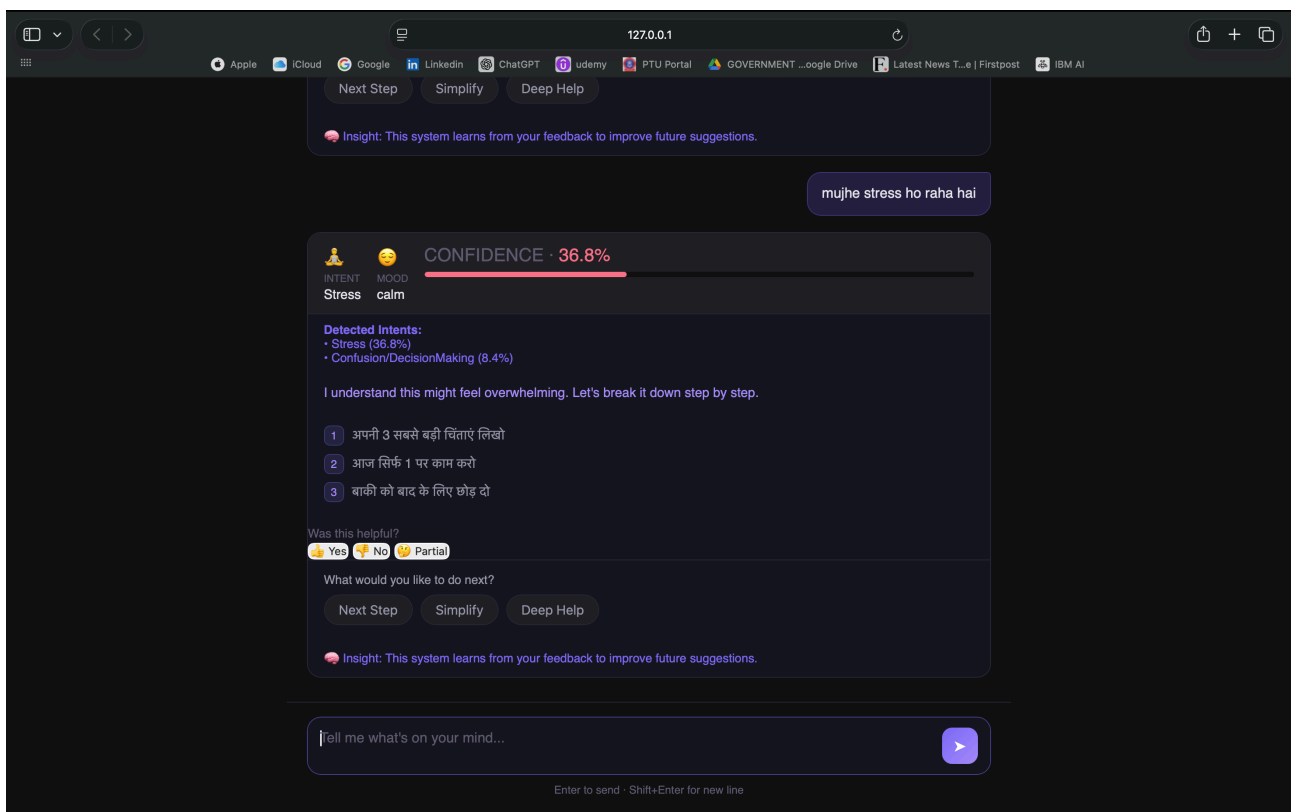
Frontend Dashboard: Main chat UI where users type messages. Input box + chat area.



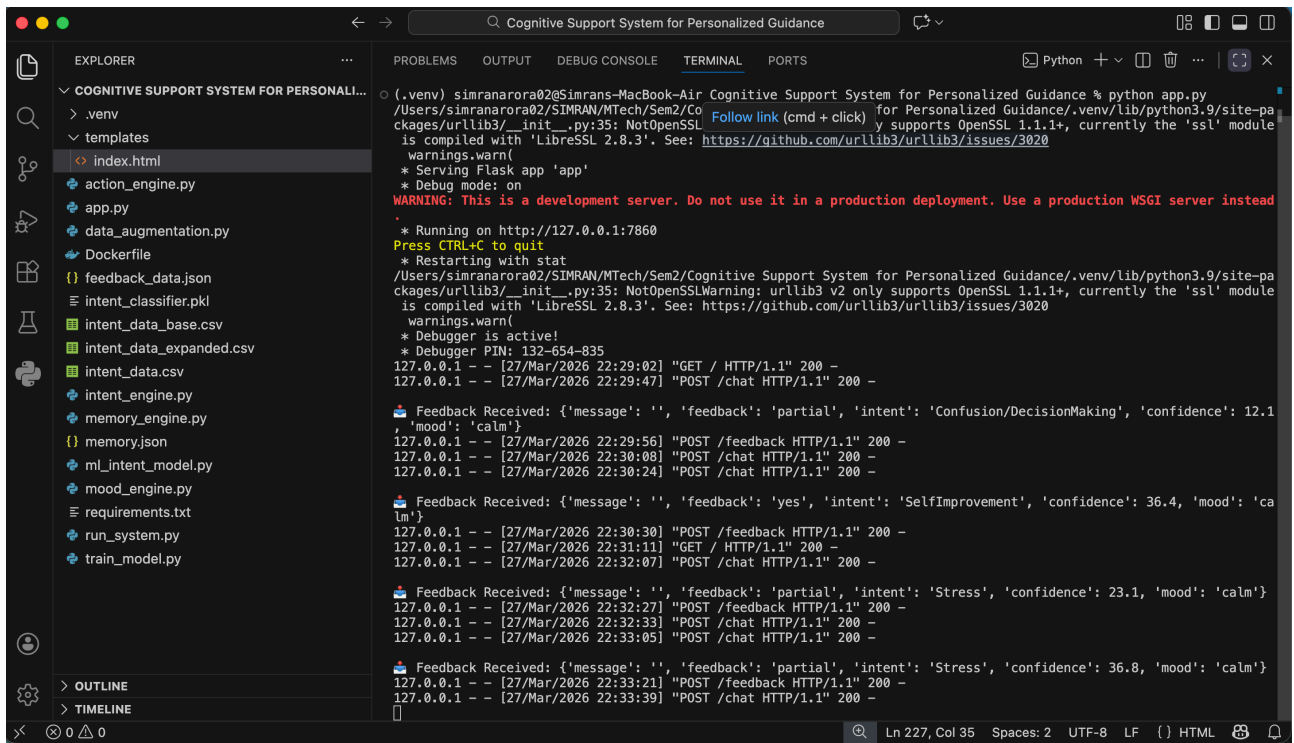
Sample Chat 1



Sample Chat when user clicks "Deep Help" button.

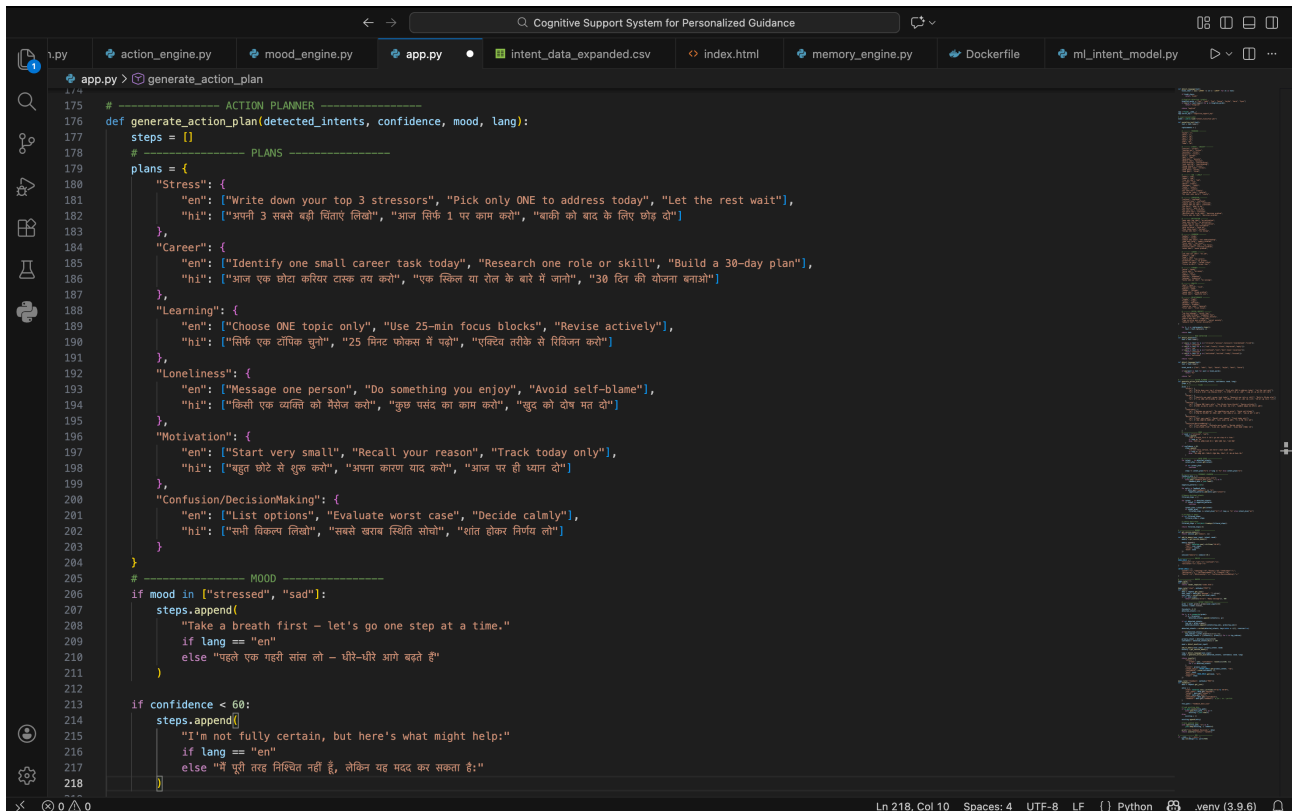


User typing “mujhe stress ho raha hai” → system response in empathetic Hindi + English



```
o (.venv) simranarora02@Simrans-MacBook-Air Cognitive Support System for Personalized Guidance % python app.py
/Users/simranarora02/SIMRAN/MTech/Sem2/Cognitive Support System for Personalized Guidance/.venv/lib/python3.9/site-pa
ckages/urllib3/_init_.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module
is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead
* Running on http://127.0.0.1:7860
Press CTRL+C to quit
* Restarting with stat
/Users/simranarora02/SIMRAN/MTech/Sem2/Cognitive Support System for Personalized Guidance/.venv/lib/python3.9/site-pa
ckages/urllib3/_init_.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.1.1+, currently the 'ssl' module
is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
* Debugger is active!
* Debugger PIN: 132-654-835
127.0.0.1 -- [27/Mar/2026 22:29:02] "GET / HTTP/1.1" 200 -
127.0.0.1 -- [27/Mar/2026 22:29:47] "POST /chat HTTP/1.1" 200 -
Feedback Received: {'message': '', 'feedback': 'partial', 'intent': 'Confusion/DecisionMaking', 'confidence': 12.1, 'mood': 'calm'}
127.0.0.1 -- [27/Mar/2026 22:29:56] "POST /feedback HTTP/1.1" 200 -
127.0.0.1 -- [27/Mar/2026 22:30:08] "POST /chat HTTP/1.1" 200 -
127.0.0.1 -- [27/Mar/2026 22:30:24] "POST /chat HTTP/1.1" 200 -
Feedback Received: {'message': '', 'feedback': 'yes', 'intent': 'SelfImprovement', 'confidence': 36.4, 'mood': 'calm'}
127.0.0.1 -- [27/Mar/2026 22:30:30] "POST /feedback HTTP/1.1" 200 -
127.0.0.1 -- [27/Mar/2026 22:31:11] "GET / HTTP/1.1" 200 -
127.0.0.1 -- [27/Mar/2026 22:32:07] "POST /chat HTTP/1.1" 200 -
Feedback Received: {'message': '', 'feedback': 'partial', 'intent': 'Stress', 'confidence': 23.1, 'mood': 'calm'}
127.0.0.1 -- [27/Mar/2026 22:32:27] "POST /feedback HTTP/1.1" 200 -
127.0.0.1 -- [27/Mar/2026 22:32:33] "POST /chat HTTP/1.1" 200 -
127.0.0.1 -- [27/Mar/2026 22:33:05] "POST /chat HTTP/1.1" 200 -
Feedback Received: {'message': '', 'feedback': 'partial', 'intent': 'Stress', 'confidence': 36.8, 'mood': 'calm'}
127.0.0.1 -- [27/Mar/2026 22:33:21] "POST /feedback HTTP/1.1" 200 -
127.0.0.1 -- [27/Mar/2026 22:33:39] "POST /chat HTTP/1.1" 200 -
[]
```

Feedback logs on the backend



```
175 # ----- ACTION PLANNER -----
176 def generate_action_plan(detected_intents, confidence, mood, lang):
177     steps = []
178     # ----- PLANS -----
179     plans = {
180         "Stress": {
181             "en": ["Write down your top 3 stressors", "Pick only ONE to address today", "Let the rest wait"],
182             "hi": ["अपनी 3 सबसे बड़ी विलक्षणताएं लिखें", "आज सिर्फ 1 पर काम करें", "बाकी को बाद के लिए छोड़ दें"]
183         },
184         "Career": {
185             "en": ["Identify one small career task today", "Research one role or skill", "Build a 30-day plan"],
186             "hi": ["आज एक छोटा करियर टास्क तय करें", "एक स्किल या रोल के बारे में जानें", "30 दिनों की योजना बनाओ"]
187         },
188         "Learning": {
189             "en": ["Choose ONE topic only", "Use 25-min focus blocks", "Revise actively"],
190             "hi": ["सिर्फ एक टॉपिक चुनें", "25 मिनट फोकस ब्लॉक में पढ़ें", "एक्टिव तरीके से रिवीजन करें"]
191         },
192         "Loneliness": {
193             "en": ["Message one person", "Do something you enjoy", "Avoid self-blame"],
194             "hi": ["किसी एक व्यक्ति को मैसेज करें", "कुछ ससंद का काम करें", "खुद को दोष मत दें"]
195         },
196         "Motivation": {
197             "en": ["Start very small", "Recall your reason", "Track today only"],
198             "hi": ["बहुत छोटे से शुरू करें", "अपना कारण याद करें", "आज पर ही ध्यान दें"]
199         },
200         "Confusion/DecisionMaking": {
201             "en": ["List options", "Evaluate worst case", "Decide calmly"],
202             "hi": ["सभी विकल्प लिखें", "सबसे खराब स्थिति सोचें", "शांत होकर निर्णय लें"]
203         }
204     }
205     # ----- MOOD -----
206     if mood in ["Stressed", "sad"]:
207         steps.append(
208             "Take a breath first - let's go one step at a time."
209             if lang == "en"
210             else "पहले एक गहरी सांस लो - धीरे-धीरे आगे बढ़ते हैं"
211         )
212     if confidence < 60:
213         steps.append(
214             "I'm not fully certain, but here's what might help:"
215             if lang == "en"
216             else "मैं पूरी तरह निश्चित नहीं हूँ, लेकिन यह मदद कर सकता है:"
217         )
218     return steps
```

Action Plan Preview

```
app.py > normalize_text
28 def normalize_text(text):
29     text = text.lower()
30
31     replacements = {}
32     # ----- PRONOUNS -----
33     "mujhe": "i",
34     "main": "i",
35     "mera": "my",
36     "meri": "my",
37     "mere": "my",
38     "hum": "we",
39     "hume": "we",
40     # ----- STRESS / ANXIETY -----
41     "tension": "stress",
42     "tension hai": "stress",
43     "preshan": "stress",
44     "pressure": "stress",
45     "bojh": "burden",
46     "darr": "fear",
47     "ghabrahat": "anxiety",
48     "ghabra raha": "anxious",
49     "overthinking": "overthinking",
50     "soch soch ke": "overthinking",
51     "dimag kharab": "stress",
52     "dimag phat raha": "stress",
53     "thak gayi": "tired",
54     "thak gayi": "tired",
55     # ----- SAD / LONELY -----
56     "dukhi": "sad",
57     "udaas": "sad",
58     "rona aa raha": "sad",
59     "ro raha": "sad",
60     "akela": "lonely",
61     "akelapan": "lonely",
62     "tanha": "lonely",
63     "tanhai": "lonely",
64     "koi nahi hai": "lonely",
65     "ignore kar rahe": "ignored",
66     "koi baat nahi karta": "lonely",
67     # ----- CONFUSION -----
68     "confuse": "confused",
69     "confused hoon": "confused",
70     "samajh nahi aa raha": "confused",
71     "samajh nahi aa rahi": "confused",
72     "kya kari": "what to do".
```

Normalizing Hindi/Hinglish for model

```
feedback_data.json > {} 6 > user_input
1 {
2   {
3     "time": "2026-02-12 17:14",
4     "user_input": "concerned about my Mtech thesis",
5     "intent": "Confusion/DecisionMaking",
6     "mood": "calm",
7     "confidence": 12.1,
8     "feedback": "no"
9   },
10  {
11     "time": "2026-02-20 13:14",
12     "user_input": "concerned about my Mtech thesis",
13     "intent": "Confusion/DecisionMaking",
14     "mood": "calm",
15     "confidence": 12.1,
16     "feedback": "yes"
17   },
18  {
19     "time": "2026-02-28 01:38",
20     "user_input": "I feel overwhelmed about upcoming exams and thesis",
21     "intent": null,
22     "mood": null,
23     "confidence": null,
24     "feedback": "helpful"
25   },
26  },
27  {
28     "time": "2026-03-14 18:04",
29     "user_input": "I feel lonely and empty",
30     "intent": "Loneliness",
31     "mood": "sad",
32     "confidence": 39.2,
33     "feedback": "yes"
34   },
35  },
36  {
37     "time": "2026-03-20 04:21",
38     "user_input": "I feel lonely and empty",
39     "intent": "Loneliness",
40     "mood": "sad",
41     "confidence": 39.2,
42     "feedback": "no"
43   },
44  },
45  {
46     "time": "2026-03-21 11:04",
47     "user_input": "I feel lonely and empty",
48     "intent": "Loneliness",
```

Backend Feedback Learning